

Power–log envelopes from a ratio limit

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2026-09-07

Abstract

First step of the mixed-ratio general- d monomial programme (Astra #15, route A). If f is bounded on $(0, \infty)$ and $f(M)/(M^{-\lambda}(\log M)^r) \rightarrow C$, then $|f(M)| \leq K M^{-\lambda}(1 + \max(0, \log M))^r$ for all $M > 0$ (`envelope_of_tendsto_ratio`); for fixed $t > 0$, $f(Nt)/(N^{-\lambda}(\log N)^r) \rightarrow Ct^{-\lambda}$ (`tendsto_scaled_ratio`); and for $N \geq e$, $0 < t \leq 1$, $|t^{q-1}f(Nt)/(N^{-\lambda}(\log N)^r)| \leq K2^r t^{q-\lambda-1}$ (`scaled_envelope_bound`). These are exactly the ingredients of the dominated-coordinate transfer lemma of the next unit. Also this unit: the independent statement-level fidelity review v2 of units 161–176 (no blocking findings; five prose qualifications applied). Zero `sorry/axiom`.

1 The things formalised

```
theorem envelope_of_tendsto_ratio (f) (l) (r) (C B) (hl : 0 < l)
  (hB : M, 0 < M → |f M| ≤ B)
  (hlim : Tendsto (fun M => f M / (M ^ (-l) * log M ^ r)) atTop ( C )) :
    K, 0 < K M, 0 < M → |f M| ≤ K * M ^ (-l) * (1 + max 0 (log M)) ^ r
theorem tendsto_scaled_ratio (f) (l) (r) (C) (hlim) (t) (ht : 0 < t) :
  Tendsto (fun N => f (N * t) / (N ^ (-l) * log N ^ r)) atTop ( (C * t ^ (-l)))
theorem scaled_envelope_bound (f) (l q) (r) (K) (hK : 0 < K) (henv) (N t)
  (hN : exp 1 N) (ht : 0 < t) (ht1 : t ≤ 1) :
    |t ^ (q - 1) * f (N * t) / (N ^ (-l) * log N ^ r)| ≤ K * 2 ^ r * t ^ (q - l - 1)
```

2 Mathematics

Beyond a threshold $M_1 = \max(M_0, 2)$ the ratio is bounded by $|C| + 1$ and $\log M \leq 1 + \max(0, \log M)$; below it, $|f| \leq B \leq BM_1^\lambda M^{-\lambda}$ since $M^{-\lambda} \geq M_1^{-\lambda}$. The scaled limit factors as $(f(Nt)/((Nt)^{-\lambda} \log(Nt)^r)) \cdot t^{-\lambda} \cdot ((\log N + \log t)/\log N)^r$. The domination uses $\max(0, \log(Nt)) \leq \log N \leq 2 \log N - 1$ so that $(1 + \max(0, \log(Nt)))^r \leq (2 \log N)^r$ once $\log N \geq 1$; it never divides by $\log(Nt)$, so $Nt < 1$, $Nt = 1$ and $r = 0$ need no case split.

3 Review v2

The independent review of units 161–176 (`tide-log/gpt6_fidelity_review_v2.md`) found no blocking or should-fix statement errors; it confirmed the Gamma/log constant (no factorial in one dimension), the monomial bridge (genuine Bochner integrability plus `ofReal` injectivity, not a lossy `toReal`), the Gaussian threshold including equality, and the mirror remark. Its five prose qualifications were applied: zero-model semantics of the quotient calculus; the $B_p^\varphi \neq 0$ qualification of the $1/\log N$ regime; the deterministic-ratio reading of the chart “posterior”; the positive-multiple scope of the Gaussian threshold; and the $\beta N > 0$ scope of the exact monomial reductions.

4 Lean notes

- Threshold $\max(M_0, 1)$ is *not* enough: at $M = 1$ Lean's totalised division makes the ratio $f(1)/0 = 0$, so the “ratio bound” carries no information; use $\max(M_0, 2)$.
- `le_or_lt` is gone; use `le_or_gt`.
- After rewriting $\log(Nt) = \log N + \log t$ in the goal, rewrite the same in the nonvanishing hypothesis, or `field_simp` cannot clear the denominator.
- `field_simp` does not normalise $\log N \cdot (\log N)^{-1} + \log t \cdot (\log N)^{-1}$ inside a power; rewrite $1 + \log t (\log N)^{-1}$ as $(\log N + \log t)/\log N$ first, then `div_pow`.